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Influence of stress-induced and thermal cracking on physical properties and microstructure of La Peyratte granite

C. David*, B. Menéndez, M. Darot

Ecole et Observatoire des Sciences de la Terre, Université Louis Pasteur — UMR CNRS 7516, 5, rue René Descartes, F-67084 Strasbourg, France

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Abstract

New data are presented on the influence of thermal cracking and stress-induced cracking on several physical properties (porosity, acoustic velocity, attenuation, electrical conductivity, permeability) of a fine-grained granite from La Peyratte, France. Two sets of samples were studied: the 'TC' set was made of samples in which thermal cracking was induced, the 'MC' set included samples in which cracks were generated at increasing stress levels in mechanical tests. Good correlations were found between the evolution of the physical properties and the amount of damage induced. For the mechanical cracking process, we show that acoustic emissions recording was the key point in order to control in real time the evolution of the cracking processes. The strong anisotropy in the acoustic velocity data found only in the MC samples is clearly due to the anisotropy of the crack distribution, as shown by microstructure analyses and 3D reconstructions of crack networks from confocal scanning laser microscopy images. A model for the propagation of elastic waves in cracked solids was used in order to infer the crack density and the mean crack aspect ratio for the granite samples. Our results show that the aspect ratio of artificially induced cracks is significantly smaller than that of cracks in 'natural' rocks. Network modelling for fluid flow in porous media was done using data from mercury injection tests and available information on the crack network geometry as input data. Although the numerical model gives the correct order of magnitude for the permeability, it could not account for relative permeability evolution with induced damage, presumably because of oversimplified assumptions in the model. All in all, our study confirms the strong control of the rock microstructure on physical properties. The data set presented in this paper might be very useful for studies on safety assessment for potential nuclear waste storage in geological environment. © 1999 Elsevier Science Ltd. All rights reserved.

1. Introduction

Most of the physical properties of crustal rocks are influenced by cracks which origin arises from various geological processes. Two of them are particularly important: thermal processes and tectonic loading. The first one is related to variations in temperature, the second one is associated with the development of high deviatoric stresses.

In recent years, thermal cracking has been extensively investigated in the laboratory, and many related papers can be found in the literature. Two important issues are concerned by thermal cracking processes: the extraction of geothermal energy and the characterization of potential nuclear waste disposal. In both situations, the problem is basically the same, i.e. how do changes in temperature influence the physical properties of rocks? Laboratory data are available which show clearly how thermal cracking modifies elastic moduli [1], permeability [2], acoustic velocities [3], attenuation [4] and mechanical strength [5] in crystalline rocks. Calleja and Ruiz de Argandoña [6] and more recently Jansen et al. [7] investigated the development of thermal cracks by monitoring continuously acoustic emissions (AE) generated during thermal cycles in a granite: the latter showed that microcracks open at temperatures above 80°C, and that microcracks can coalesce to form a macroscopic fracture. This may

* Corresponding author. Tel.: +33-(0)388-416-481; fax: +33-(0)388-616-747.

E-mail address: christian.david@eost-ulp.u-strasbg.fr (C. David)

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have dramatic consequences on the permeability evolution of a nuclear waste disposal. SEM analysis coupled with stereological analysis [5] and mercury injection tests [8] have proven to be useful tools to characterize thermally-induced cracks in granites. Such data are useful in models for the development of thermal cracking in crystalline rocks [9,10].

It is well known that intense cracking occurs before a rock fails under deviatoric stresses. Surprisingly, only few studies have focused on the analysis of the crack distribution which develops during a mechanical test. Laboratory studies have shown how the physical properties vary while increasing compressive stresses are applied to rocks: among others, variations in permeability [11], acoustic velocity [12] and attenuation [13] in granites have been reported. One of the major contributions of these studies was to show that microcracking and dilatancy affect significantly the rock properties when the stress level is larger than approximately half the ultimate strength of the rock, in agreement with basic knowledge in rock mechanics [14]. Dilatancy was the subject of detailed studies [15]. These experiments were generally conducted until the rock samples fail by fracture localization: obviously at this stage, it is not possible anymore to investigate the origin of the changes observed in the prefailure region, that is the stress-induced crack distribution. Such a quantitative analysis of crack distributions would be extremely useful for testing available theoretical models on the evolution of rock properties with in situ conditions. To study the microstructural properties of stress-induced cracks, one has to perform mechanical tests at stress levels lower than the rock strength. This was done for example by Wong [16], following the pioneering work of Tapponnier and Brace [17]. Using SEM, these authors investigated the relationships among cracking, mineralogy and grain size in Westerly granite samples deformed in triaxial experiments under a wide range of temperature and pressure conditions.

Our goal in this work was to investigate how cracking influences some important physical properties of granites, and more specifically to study the effect of stress-induced cracking compared to thermal cracking. For this purpose, two sets of granite samples were prepared, one, the 'TC' set (for 'thermal cracking'), in which thermal cracks were induced using classical methods and a second one, the 'MC' set (for ' mechanical cracking') in which cracks were generated under increasing stress conditions in triaxial tests. For both sets, several properties were measured: acoustic velocities, *P*-wave attenuation, permeability and electrical conductivity. In order to avoid possible problems related to poor reproducibility of the measurements from sample to sample, all the measurements were first done on each rock sample before cracking was induced: doing so, we have a precise knowledge of the effect of cracking on each individual sample. After completing this stage, a quantitative analysis of the crack distributions was done on the cracked samples. The resulting characteristics of the crack distributions greatly helped in the interpretation of the experimental results.

2. Description of rock material and 'damaging' procedures

2.1. Starting material

The selected rock is a fine grained granite from La Peyratte, France. The grain size for La Peyratte granite ranges from about 0.5 to 1.5 mm, and the mineralogical composition is 40% plagioclase, 26% K-felspar, 24% quartz and 10% mica [18]. From a single block, core samples were prepared with both flat surfaces ground squared: nominal length and diameter for the specimens were, respectively, 40 and 20 mm. The choice for this fine grained granite was imposed by the small sample size required by our experimental devices. Transport properties of La Peyratte granite were studied in previous works [2,19].

2.2. Preparation of the thermal cracking (TC) set of samples

Granite samples were heated in a furnace at a low heating rate (1°C per min) until a nominal temperature was reached. Temperature was then held constant during an hour, then the system was cooled down at the same low rate, in order to avoid thermal shock during the heating/cooling cycle. Following this procedure, thermal cracking develops efficiently provided the ultimate heating temperature is high enough [2].

2.3. Preparation of the mechanical cracking (MC) set of samples

Other core samples were subjected to high deviatoric stresses in a conventional triaxial apparatus in such a way that irreversible damage was induced. As we were not interested in a thorough study of the mechanical behavior of La Peyratte granite, we arbitrarily fixed the confining pressure to a low value ($\sigma_3 = 10$ MPa). The mechanical tests were conducted at room temperature and under a constant strain rate $\dot{\varepsilon} \cong 2 \times 10^{-5}$ s^{-1}: with such low P–T conditions, we are sure that the overall behavior of the granite is brittle. For more details on the experimental device, one can refer to Zhang et al. [20] and David et al. [21]. We followed the procedure used successfully by Wong [16] on Westerly granite and more recently by Menéndez et al. [22] in a study on the micromechanics of faulting in

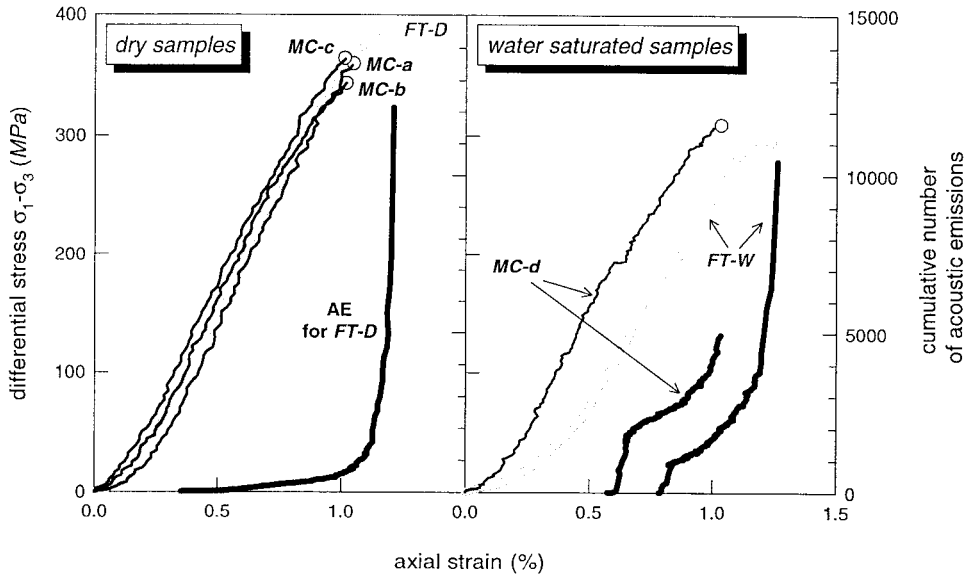


Fig. 1. Axial strain–differential stress plots. Complete failure tests (FT) are shown in thick grey for dry (left) and water saturated (right) samples. Open circles give the maximum stress experienced by the MC samples. Thick black lines correspond to acoustic emission counting for selected samples.

Berea sandstone. Selected rock samples were loaded up to a given stress level close to (but lower than) the ultimate strength of the material, and then unloaded. Doing so, one can reasonably hope that the higher the stress level experienced by the sample, the larger the amount of irreversible damage induced. However, this is not systematically true because even for rock samples cored from a same block the mechanical behavior is never exactly the same. Therefore, this raises the following question: which relevant criterion can be used to estimate in real time during the loading stage the amount of damage induced? In his experiments, Wong [16] loaded his rock samples up to given values of stress: doing so he relied on both the reproducibility in sample strength (and more generally in mechanical behavior) and on his skills to control manually the loading machine when getting close to the peak stress. Obviously, one would prefer an automated and more systematic 'damaging' technique for such studies: we will show that monitoring acoustic emissions (AE) while deforming the rock samples provided valuable information to allow a fair control of the experiment. Piezoelectric transducers with resonant frequency of 1 MHz coupled with a discriminator device were used to pick up the onset of small brittle events inside the rock samples. Monitoring acoustic emissions is useful in several aspects: to detect the closeness of failure [23], to locate microfracturing events during triaxial loading [24] and to determine the onset of yielding and shear-enhanced compaction in high porosity rocks [20,21]. To estimate in our study the amount of stress-induced damage experienced by each granite sample, we used the cumulative number of AE. As this parameter was displayed in real time during the mechanical test, it could be used to control the experiment in such a way that a given specimen experienced more or less damage compared to previously tested samples. In a first approximation, it is reasonable to assess that the more acoustic emissions recorded, the more damage induced in the rock sample. Damage can result either from generation of new cracks, or from propagation of existing cracks. The key point in our procedure is that the operator can use an objective criterion based on AE counting to interrupt the loading. As a preliminary step, it is necessary to conduct a complete triaxial experiment until sample failure occur: such a failure test provides an estimate for the expected upper limit of AE counting before fracture localization. This upper limit is used to scale the damage level for the preparation of the MC samples.

3. Description of the 'TC' and 'MC' sample sets

The TC set consists in two core samples, corresponding, respectively, to plateau temperatures of 220 and 450°C. As expected, more thermal cracking was induced at higher heating temperatures. These samples will be referred to hereafter as TC-220 and TC-450, respectively. Four samples were available in the MC set, with references MC-a, MC-b, MC-c and MC-d. This classification corresponds to increasing order of induced damage. We are conscious of the fact that we are working on a limited set of samples, which is not recommended to get statistically meaningful results. Ideally, one would need to study 'many' samples for

Table 1
Results from mechanical tests on La Peyratte granite samples. NA means nonapplicable

Sample	Initial porosity (%)	Young's modulus (GPa)	Differential stress at failure (MPa)	Maximum differential stress (MPa)	Maximum AE counts (cumulative)	Comments
FT-D	–	41	387	NA	12000	complete triaxial on a dry sample
MC-a	0.40	43	NA	360	1300	dry
MC-b	0.41	41	NA	344	2800	dry
MC-c	0.46	43	NA	365	4600	dry
FT-W	0.52	40	302	NA	10500	complete triaxial on a water saturated sample
MC-d	0.47	42	NA	312	6800	water-saturated

each damage state. This was not possible for practical reasons (preparation and measurements are very time-consuming). Sample MC-d was prepared in a different way from the others: it was saturated with water and deformed under undrained conditions. The reason for conducting an undrained test was to increase the efficiency of the cracking process during the loading cycle: this point will be discussed later. Two failure tests were conducted, respectively on a dry sample (sample FT-D, for 'failure test — dry') and on a water saturated sample under undrained conditions (FT-W, for 'failure test — wet'). Both failure tests were used as master curves for calibrating the mechanical behavior of La Peyratte granite. In Fig. 1, the differential stress–axial strain curves are plotted for all MC samples. Table 1 gives a summary of the important features inferred from the mechanical tests. Let us comment these results separately for the dry and water-saturated core samples.

For the dry samples, the first striking result is the fairly good reproducibility in the loading paths. This is confirmed by the small scatter in Young's moduli calculated from the mean slope in the linear parts of the plots (Table 1). Sample FT-D failed at 387 MPa differential stress, which falls within the range of compressive strengths for granites in Lockner's recent compilation [25]. The maximum stress experienced by the MC samples corresponds to more than 94% of this value. If we compare samples MC-b and MC-c, we can see that the cumulative number of acoustic emissions recorded does not correlate well with the maximum stress level (Table 1), whereas there is a good correlation between AE counts and the amount of damage induced, as we will see later. If we assume that induced damage is related to how close the stress level is to the true sample strength, this shows that the strength is not very reproducible, on the contrary to elastic moduli. A possible explanation for the scatter in rock strength is heterogeneity from sample to sample, which affects more the mechanical strength than elastic properties.

If we focus now on the saturated samples, both curves do not overlap as well as for the dry samples: reproducibility is weaker. However, the shapes of the curves are similar for stresses larger than about 100 MPa: the observed discrepancy is mainly due to the behavior in the low stress region. A possible explanation is that the initial crack content was larger in sample FT-W, so that axial strain was enhanced due to crack closure at low stresses: the fact that sample FT-W had the highest porosity (Table 1) supports this idea. The second important feature is that the strength is about 80% lower than for the dry samples. This can be explained if we take into account the dependence of rock strength with effective confining pressure. Because the mechanical tests are conducted under undrained conditions, the water can not escape from the specimen and fluid pressure builds up accompanying compressive loading. The effective confining pressure is lower and consequently the rock strength is expected to be smaller than for the dry rock [26]. Unfortunately the experimental set-up did not allow for the measurement of pore pressure during the loading of the wet samples. Finally, another striking result is the presence of an instability in both loading curves at differential stresses close to 190 MPa, corresponding approximately to 60% of the peak stress, where dilatancy is expected to start for a rock under compressive load [27]. The instability is characterized by an instantaneous acceleration in axial strain, a sharp increase in acoustic emissions counting and a decrease of the mean slope above the threshold stress. The sudden acceleration in AE activity not found in the dry tests shows clearly that the instability is associated with brittle processes and with the presence of water. Our interpretation is as follows. Below the threshold stress, water seems to inhibit elastic energy release: no acoustic emissions were recorded below the threshold stress, whereas in the dry samples the number of AE recorded increased continuously starting from a lower stress level. The early AE in the dry samples are presumably associated with the growth of preexisting cracks rather

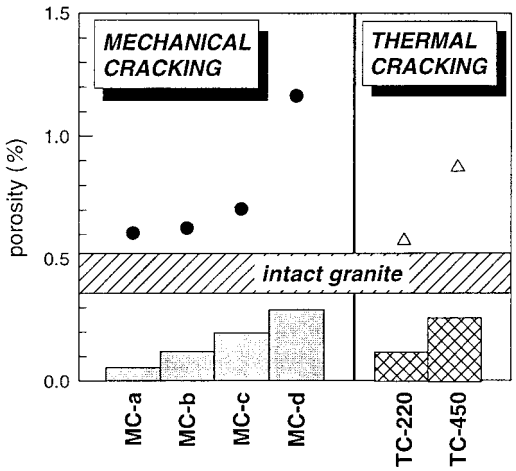


Fig. 2. Porosity of MC samples (solid circles), TC samples (open triangles) and range for intact granite (horizontal bar). The height of the vertical bars scales as the maximum number of acoustic emissions recorded in the mechanical tests (see values in Table 1) and the maximum temperature experienced by the heated samples.

than the creation of new cracks. When cracks are filled with water, no AE are detected in the early stage because less elastic energy is released due to the lubricating effect of water. It is well known that water reduces surface energy and enhances slow crack growth at low stresses [28]. Due to the orientation of the applied stress, cracks oriented perpendicular to σ_1 will tend to close preferentially, and water will be expelled from them towards other cracks. Some of these subhorizontal cracks may become isolated from the connected crack network: in such cracks, further pore pressure build-up will inhibit further crack closure. At the threshold stress level, the dilatancy point was reached, new cracks are generated and acoustic emissions are recorded. As soon as the newly formed cracks connect to the fluid filled crack network, they are instantaneously filled with water, and the overall pore pressure decreases sharply. Both pore pressure variations and water invasion in the created cracks will enhance crack propagation and acoustic emission activity. Therefore, the threshold stress acts as a percolation threshold for the AE activity, in agreement with our observations. In addition, fluid pressure release in subhorizontal cracks gives rise to sudden increase of the effective stress acting on them: as a consequence, these cracks will tend to close almost instantaneously. Such a mechanism is supported by the accelerated axial strain increase observed at the threshold stress.

4. Physical properties of the granite samples

Laboratory measurements conducted on our set of samples are: porosity, *P*-wave velocity, *P*-wave attenuation, permeability and electrical conductivity. The measurements were done first on a suite of eight granite samples (among them the MC samples prior to mechanical testing) which did not experience any cracking procedure: this data set will be called the 'intact rock' data set. The mean values and experimental scatter for the properties of the intact granite are well defined and provide a good comparison basis to study the variation in rock properties for the sets of deformed samples.

4.1. Porosity

Porosity was classically inferred from measuring successively the weight of dry, water saturated and water immersed samples. The results are reported in Fig. 2 and Table 2. The porosity for the intact granite ranges between 0.36 and 0.52% (horizontal bar in Fig. 2). All the deformed samples have a larger porosity than the intact rock. Interestingly, the measured values are comparable for the TC and the MC set. In addition, a clear correlation was found between porosity and on one hand the maximum temperature experienced by the TC samples, and on the other hand the maximum number of acoustic emissions recorded for the MC samples (vertical bars in Fig. 2). This shows in particular that AE counts is a good parameter to characterize in first approximation the damage induced in the mechanical tests. Obviously we have not enough data to test the reproducibility of the trend observed for the AE counts vs. porosity increase The sharp porosity increase for MC-d shows that conducting a triaxial test on a water saturated sample under undrained conditions is very efficient in creating new crack porosity.

Table 2
Measured physical properties for intact and deformed granite samples

Sample	Range for intact rock	MC-a	MC-b	MC-c	MC-d	TC-220	TC-450
Porosity (%)	0.35–0.52	0.61	0.62	0.70	1.20	0.57	0.87
Axial *P*-wave velocity (m/s)	5450–5540	5283	5237	5168	5025	4836	3729
Velocity anisotropy factor (%)	0.1–5.3	21	29	28	45	3.8	2.8
Attenuation factor $1000/Q$	7–12	43	48	42	83	43	40
Formation factor	5000–7000	3574	2178	2742	561	2459	614
Permeability at $P_{eff}=3$ MPa (10^{-21} m^2)	6–32	384	656	1100	392	678	1570

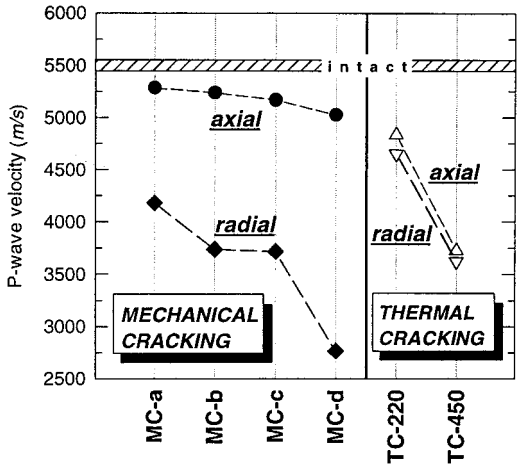


Fig. 3. *P*-wave velocity data for the intact and deformed samples. Both axial and radial velocity are plotted, corresponding to wave propagation respectively parallel and perpendicular to the core sample axis.

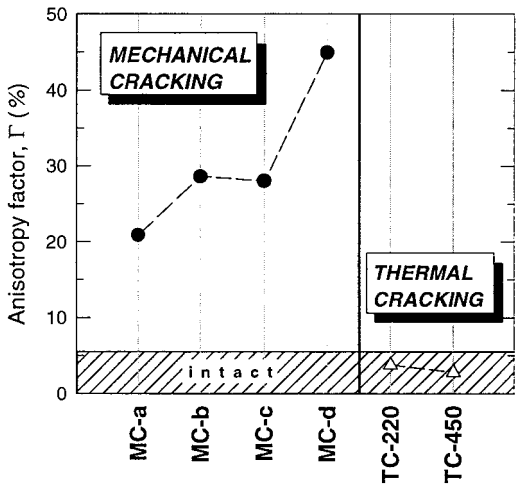


Fig. 4. Anisotropy factor Γ for *P*-wave velocities measured in orthogonal directions (see text for details).

4.2. P-wave velocity

The velocity of compressional waves were measured at room conditions using the ultrasonic pulse transmission technique. The experimental set-up includes a waveform generator, two piezoelectric transducers mounted on sample holders, a small loading frame and a digital oscilloscope. The signal sent to the transmitter is a single sinusoidal period with 700 kHz dominant frequency which matches the resonance frequency of the transducers. Velocity is calculated from the travel time of the *P*-waves through the rock sample. A contact force of 600 N (measured by a load cell) was systematically applied to insure a tight contact between the rock specimen and the sample holders. In addition, the quality of the recorded signals was largely improved by spreading a gel onto the contact surfaces of the sample holders.

P-wave velocities were measured along three orthogonal directions: in the axial direction (axial velocity V_{ax}) and across two orthogonal diameters from which a mean radial velocity is inferred ($\bar{V}_{rad}$). This was done in order to check for the existence of anisotropy for the acoustic properties. The velocity data are reported in Table 2 and plotted in Fig. 3. As usual, the horizontal bar gives the range of velocities for the intact granite samples. *P*-wave velocities measured on the deformed samples are systematically lower compared to the intact rock. As expected, there is a negative correlation between velocity and induced damage for both TC and MC sets. This was observed in studies on thermal cracking in a granite [3] and in real time experiments on a granite loaded under uniaxial stress conditions up to failure [13]. Whereas comparable values of porosity were found for both sets (Fig. 2), the axial velocity decrease for the TC samples is much sharper than that observed for the MC samples. A possible explanation for this interesting observation is that the geometry of the cracks (i.e. size, shape) are very different, depending on the cracking process. Such a behavior can also result from different crack orientation distributions. Without ruling out the first argument on crack geometry variations, we will show that our data set strongly supports the assumption of anisotropy.

4.3. Acoustic velocity anisotropy

It is well known that stress-induced cracks in a triaxial test have a preferential orientation parallel to the direction of maximum principal stress (i.e. along the core sample axis). On the opposite, we expect thermal stresses to induce a more or less isotropic crack distribution. Indeed La Peyratte granite is homogeneous in composition and does not show any crystal preferred orientations (i.e. no particular shape fabric). Thus we expect the crack populations to be anisotropic in MC samples, and fairly isotropic in TC samples. To check on the existence of anisotropy, we compared axial to radial velocities in Fig. 3. Whereas no significant differences are observed for the thermally cracked samples, we found huge variations for the MC samples. We also calculated the so-called 'anisotropy factor':

$$\Gamma = \frac{V_{ax} - \bar{V}_{rad}}{V_{ax}} \tag{1}$$

and plotted the results in Fig. 4. For the intact granite, there is no significant anisotropy as Γ is always smaller than 5%. The MC samples have a very different behavior, with anisotropy factors larger than 21% up to 45%. This shows that a strong *P*-wave velocity aniso-

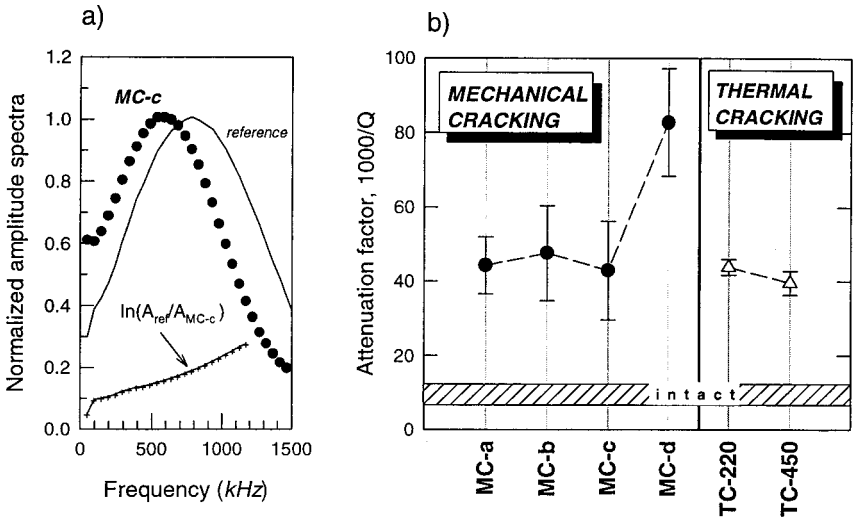


Fig. 5. Results for *P*-wave attenuation measurements in the granite samples. (a) Bell-shaped curves showing the normalized amplitude spectra for reference sample (ceramic) and deformed sample MC-c. The quality factor Q is derived from the slope of the lower curve. (b) Attenuation $1000/Q$ for intact (horizontal bar) and deformed granite samples.

tropy exists in the MC samples, and this is associated to anisotropic crack distribution in the deformed granite. Interestingly, the anisotropy factors for the TC samples fall within the range of values for intact rock samples: therefore, thermal cracking did not result in measurable anisotropy in these samples, at least from the acoustic properties viewpoint. The variation in velocity decrease between the TC and the MC samples (Fig. 3) is also supported by the existence of crack anisotropy. As *P*-waves propagate in the axial direction, compressional motion is polarized in a direction parallel to the core axis: because of the preferred orientation of stress-induced cracks, the travel times for *P*-waves propagating along the core axis direction will be less altered by new cracks generation in the MC samples than if *P*-waves would propagate in a medium where the created cracks have all possible orientations like in the TC samples. Therefore, we expect a larger velocity decrease for the latter; and this is exactly what we observed. Interestingly, the overall radial velocity decrease for the MC samples (Fig. 3) compares well with that of the TC samples, in agreement with the previous discussion.

4.4. P-wave attenuation

We used the ultrasonic pulse transmission technique described by Johnston and Toksoz [29] to measure the *P*-wave attenuation. Quality factor Q is derived from spectral amplitude ratios by comparing the amplitude spectrum of the rock sample to that of a reference sample with the same geometry, made of a material with negligible attenuation. It is important to note that this technique provides relative values, which are good estimates of the true quality factor only if the reference material has a well known Q, or if its attenuation can be considered to be virtually zero. In our case, this point is not critical because our goal is to compare the attenuation from sample to sample, and this can be achieved by using the same reference material (a non-porous ceramic) throughout the experiments. As an example, Fig. 5a shows the amplitude spectra for the reference sample and deformed sample MC-c, normalized to the respective peak amplitude. Due to attenuation in the cracked sample, the bell-shaped spectrum is shifted towards low frequencies. Plotting the logarithm of the amplitude spectra ratio vs. frequency, it can be shown [29] that the slope ψ is related to the quality factor Q by

$$Q = \frac{\pi L}{V\Psi}, \tag{2}$$

where L is the sample length and V is the axial *P*-wave velocity of the sample (here zero attenuation is assumed for the reference material). In Fig. 5a, a linear trend is effectively observed in the frequency range corresponding approximately to the bandwidth of the input waveform. Our results for intact and deformed granite samples are plotted in Fig. 5b, where attenuation is quantified by $1000/Q$. Despite of the large error bars due to poor linearity observed on the spectral ratio plots, it is clear that attenuation for the cracked samples is significantly higher than for the intact granite. This is not surprising as cracks are involved in various dissipation mechanisms, like friction or scattering [30]. On the other hand, the variation observed for the MC samples is small compared to the experimental error bars, except for sample MC-d in which there is a sharp increase in attenuation. Taking into account these errors, one can argue that

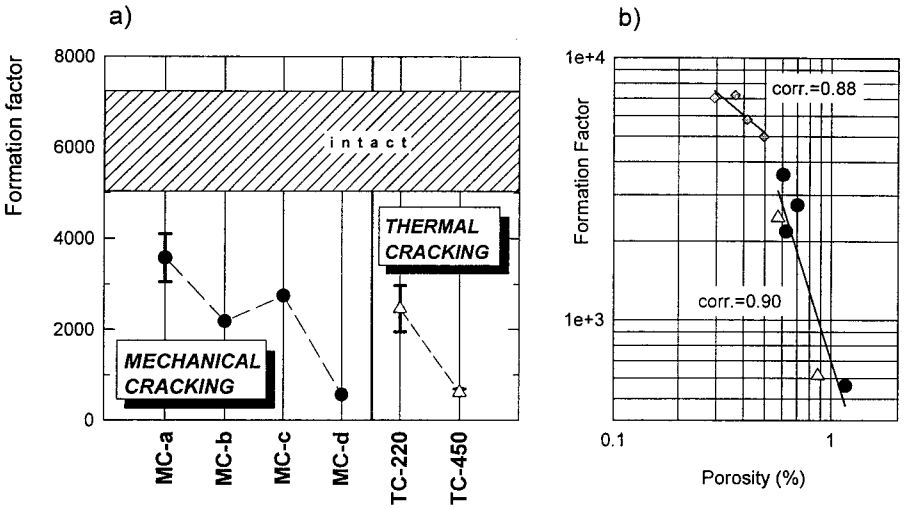


Fig. 6. (a) Formation factor F derived from electrical conductivity measurements. (b) Logarithmic plot F vs. ϕ.

there is an overall positive correlation between attenuation and induced mechanical damage. Attenuation factors for the thermal cracking set are in the same range. However, in this case there is a negative correlation between attenuation and induced thermal damage. This may be a drawback due to the fact that sample TC-450 is the only one with a length significantly different from that of the reference sample. On the other hand, such a behavior was also observed by Johnston and Toksoz [4] on thermally heated rocks. For samples heated at temperatures lower than 400°C, porosity increased, acoustic velocity and attenuation decreased with maximum temperature, like in our experiments. At 500 and 600°C, attenuation increased. To explain such a behavior, the authors argued that as rock samples are heated there is an interplay between two competing processes: cracking, which is expected to increase attenuation and outgassing of crack surfaces, which is an effective mechanism for decreasing attenuation (see the review on attenuation mechanisms by Bourbié et al. [31]). The first mechanism is operative at higher temperatures, the second one is dominant at lower temperatures. Our data agree with Johnston and Toksoz' observations [4], but we should be cautious due to the poor quality of our data. In their experiments on thermally cracked Westerly and Charcoal granites, Bauer and Johnson [3] found little change in attenuation below the temperature $T_{\alpha\beta}$ for quartz αβ transition, with Q factors close to 180 below $T_{\alpha\beta}$ down to 10 above $T_{\alpha\beta}$. Interestingly, our results for La Peyratte granite fall in the same range: this gives us some confidence in our calculated values of Q factors.

4.5. Electrical conductivity

To determine accurately the electrical conductivity of the granite samples, a series of electrical measurements were performed, following the procedure described in Le Ravalec et al. [32]. The samples were first jacketed tightly with rubber tape and saturated under vacuum with a degassed sodium chloride brine with known conductivity σ_f. After leaving the samples in the solution for several days, electrodes were applied to both ends of the samples and the effective conductivity σ_r of the brine saturated rock was measured using a fixed frequency conductivimeter at room conditions. This procedure was applied successively for the samples saturated with NaCl brines with conductivity 1, 3 and 6 S/m. Between each series of measurements with different brines, the samples were cleaned during several days by flushing deionized water through them in order to remove the salt remaining in the samples. Plotting the effective rock conductivity vs. the brine conductivity, the formation factor F was inferred from the inverse slope of the best straight line which fits the data, according to the relation $\sigma_r = \sigma_f/F + \sigma_s$, where σ_s is the so-called surface conductivity [33]. In general, the $\sigma_r = f(\sigma_f)$ plots were reasonably linear except for two samples for which large error bars were found. In Fig. 6a, one can see that the formation factor of the deformed samples is systematically lower than that of the intact granite: such a result is expected because the transport of ions is enhanced by an increase in porosity and crack density, resulting in lower values for F. There is also an overall decreasing trend for the formation factor evolution with induced damage: this is clearly the case for the TC samples, whereas a more peculiar behavior is observed for the MC samples. In Fig. 6b we plotted the formation factor vs. porosity in a log–log scale and separated the intact, MC and TC samples: such a plot allows to check on the existence of Archie type of laws. As a matter of fact, one can easily find a fairly good linear correlation for the intact granite data set, and a

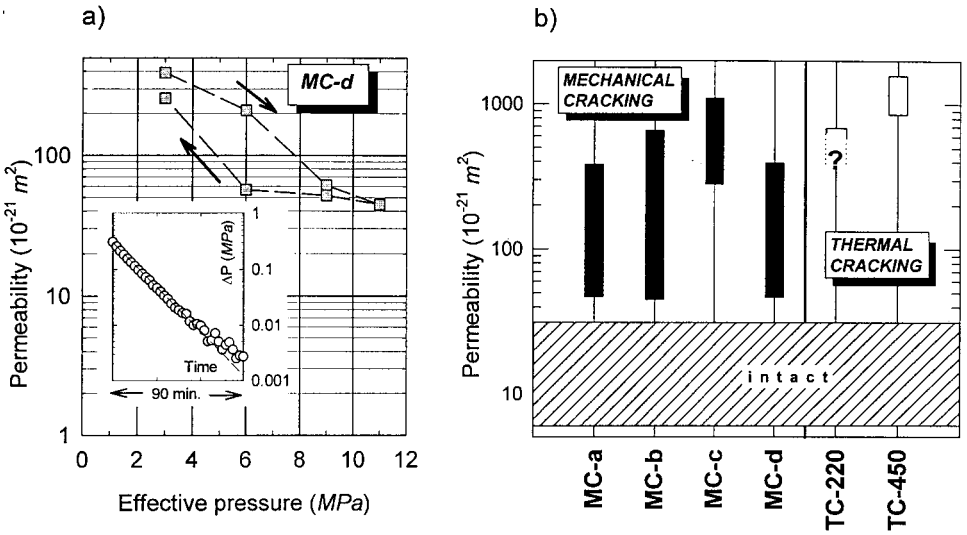


Fig. 7. (a) Permeability evolution during an hydrostatic pressure cycle on sample MC-d. The inserted plot shows the exponential pore pressure decrease during a permeability measurement using the pulse technique. (b) Permeability data for the granite samples. The length of the vertical bar gives the pressure sensitivity of permeability between 3 MPa (top) and 10 MPa (bottom) effective pressure.

weaker but consistent linear decreasing trend for the data set corresponding to all the deformed samples. Given these evolutions, we calculated two values for the m exponent and the a factor of the power law $F = a\phi^{-m}$ and found, respectively, $a = 128$ and $m = 0.7$ for the intact samples, and $a = 0.0032$ and $m = 2.7$ for the TC and MC samples taken together. This interesting result shows that intact and artificially cracked granite samples behave differently, which raises the following question: is it relevant to compare rocks in which cracking was induced by natural processes, to rocks in which man-induced cracking was imposed? Our electrical data seem to show that no single law can explain the evolution of formation factor as a function of porosity. One should mention however that we are looking here for correlations over an extremely small range of porosity. This may explain why we found m exponents out of the usual range for this parameter (between 1.3 and 2.5 [34]). Low Archie exponents were found in laboratory experiments by Le Ravalec et al. [32] and Ruffet [19] on thermally cracked mylonite ($m = 1.4$) and granites ($m \approx 1$), respectively

4.6. Permeability

Unlike all other experiments which were conducted at room conditions, permeability was measured as a function of applied effective pressure, using the experimental set-up described in David et al. [35]. Water saturated samples were covered with a Viton jacket and put into a pressure vessel with maximum capability 100 MPa. The system allows to control precisely the confining pressure in the vessel, the fluid pressure in the upstream reservoir (P_{up}) and in the downstream reservoir (P_{down}) connected to both end surfaces of the sample. As the granite samples were supposed to be used later for further analyses, we decided to work at very low effective pressures (between 3 and 10 MPa) in order to make sure that only elastic deformation was induced during a loading cycle and thus to avoid any irreversible damage to the crack distribution. To measure the permeability, we used the transient pulse method [36]: starting from a static situation where fluid pressure is homogeneous in the whole system (i.e. sample + upstream and downstream reservoirs), a small increment of pressure is applied instantaneously in the upstream reservoir and the pressure evolution with time is recorded while the pressure pulse diffuses through the sample. The amplitude of the pulse is always small (0.3 MPa) compared to the static fluid pressure (10 MPa) so that effective pressure can be considered to be constant during the test. The theory states that, under some conditions (fulfilled for our granite samples), the time-dependent pressure decrease is exponential, according to the following relation:

$$P_{up} - P_{down} = \Delta P_0 \times \exp(-\lambda t) \text{ with}$$
$$\lambda = \frac{kA}{\eta L}\left(\frac{1}{C_{up}} + \frac{1}{C_{down}}\right), \quad (3)$$

where A and L are the sample cross-section and length, respectively, η the fluid viscosity, and C_{up} and C_{down} are the compressive storage of the upstream and downstream reservoirs [37]. Given $C_{up} = 20.6 \times 10^{-9}$ m^3/MPa and $C_{down} = 5.8 \times 10^{-9}$ m^3/MPa, the sample

permeability is derived from the pressure decay plot which should be linear on a semi-log plot (see inserted graph in Fig. 7a). We plotted in Fig. 7a the permeability decrease for sample MC-c as a function of applied effective pressure. During the loading stage, the permeability evolution was found to be exponential for all our samples, in good agreement with data reported by Le Ravalec et al. [32] and with the compilation done by David et al. [21]. Fig. 7b shows how the permeability changes with induced damage in our granite samples. The upper (respectively lower) limit of the vertical bars gives the value of the permeability measured at 3 MPa (respectively 10 MPa) effective pressure: in other words, the length of the bars is a measure of the pressure sensitivity of permeability. First, let us point out that we have not a complete data set for sample TC-220, because of technical problems (leaks) during the measurements. As expected, the TC and MC samples have a higher permeability than the intact granite. Notice that for the latter, the horizontal bar gives the range of permeability measured at a fixed effective pressure (3 MPa), and that there is a much larger scatter (about a factor 5) for the permeability of the intact granite compared to other properties. This may be associated to the fact that a small heterogeneity in the crack distribution from sample to sample can induce large variation in permeability. For the thermally cracked samples, the permeability at 3 MPa effective pressure increases with increasing crack porosity. This is also true for samples MC-a–c. Surprisingly, sample MC-d which is the most damaged sample has a lower permeability than sample MC-c. At this point we have no clear explanation for such a puzzling behavior. No systematic trend was observed for the pressure sensitivity of permeability in our experiments (Fig. 7b). Darot et al. [2] reported values close to 70 nDarcies for gas permeability measurements on La Peyratte granite at low effective pressure, in reasonable agreement with our experimental data.

For both transport properties (electrical conductivity and permeability), we found that unlike for the acoustic properties, the range of measured values for the intact granite samples is quite large, and moreover is comparable to the variations observed for the deformed samples. This unexpected result may be linked to the fact that transport properties are extremely sensitive to the geometry and topology of the crack networks which can be significantly different even for samples with similar porosity values. The bulk porosity (which is directly related to induced crack damage) is not a good controlling parameter for the transport properties: more geometrical information is needed at the microcrack scale, which can be obtained by a thorough microstructure analysis.

5. Analysis of microstructures

Once all the physical measurements were done, the samples were prepared for microstructure studies. Each sample was cut into three pieces: one was kept for future use on Wood's metal injection, a second one was used for mercury injection tests and a third one was injected under vacuum with an epoxy doped with a fluorescent dye for microscope studies. For the latter, the epoxy was forced to enter into the pore space by applying a 10-MPa back pressure. For the microscope studies, we decided to use a powerful technique, confocal scanning laser microscopy (CSLM), to analyze and image the crack networks in the deformed samples. This technique has been widely used in biological sciences but only recently in Earth sciences. For a description of CSLM techniques applied to geomaterials, one can refer to Fredrich et al. [38]. There are two major advantages in using CSLM: first the high resolution in the definition of the focal plane allows to conduct stereological analysis on two-dimensional sections with a better accuracy, second this technique allows to make nondestructive serial sectioning of the samples, from which the three-dimensional structure of the crack network can be reconstructed. Let us summarize the main results of the microstructure analysis [39]. A statistical analysis on the crack orientation distribution in the deformed and intact granite samples showed that all the MC samples are strongly anisotropic with a preferred orientation of the cracks in a direction parallel to σ_1, the maximum compressive stress applied during the mechanical tests. No significant anisotropy was detected for the intact and TC samples. The results of the microstructure analysis are very helpful for the interpretation of our data on the velocity anisotropy in the granite samples. Two 3D reconstructions were also done [39] which showed the striking difference between crack patterns found in TC samples compared to MC samples. One of them, relative to sample

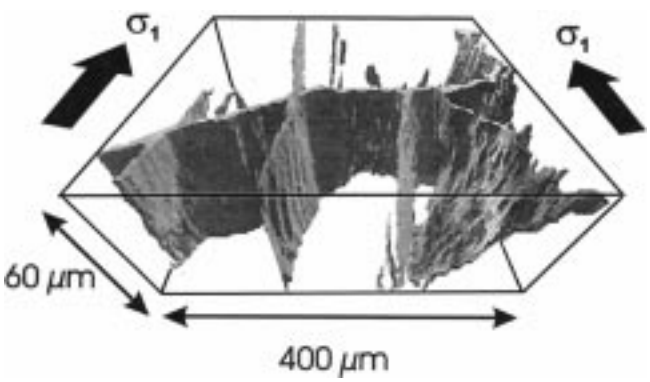


Fig. 8. Three-dimensional reconstruction of part of the crack network in sample MC-c. The arrows give the direction of application of the maximum compressive stress during the mechanical test.

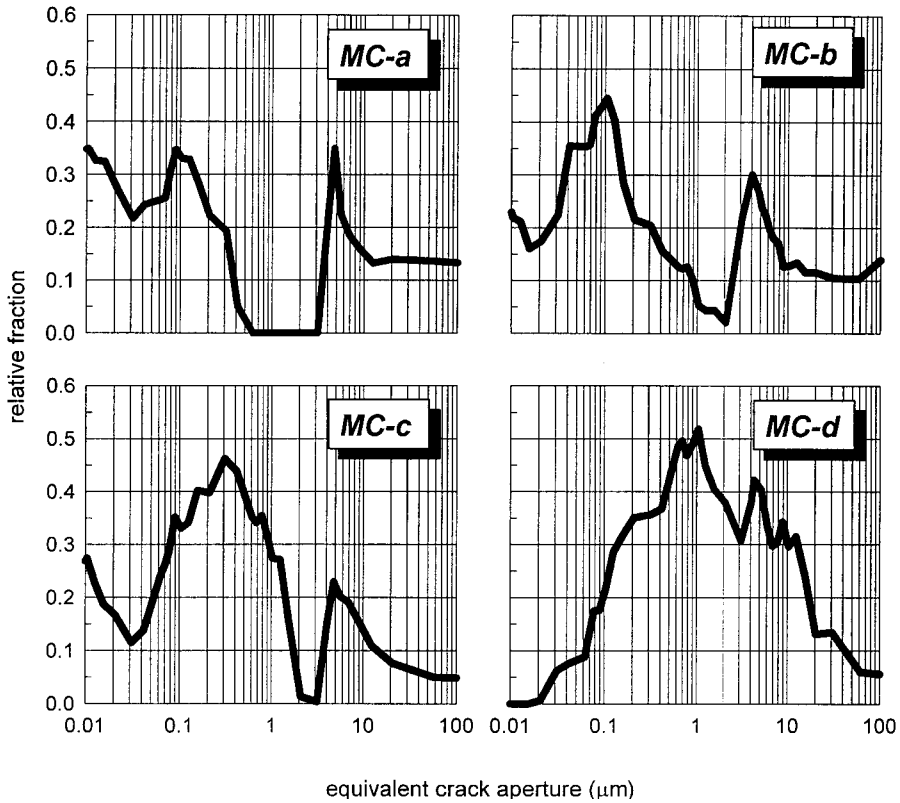


Fig. 9. Equivalent crack aperture distribution inferred from mercury injection tests on the MC samples.

MC-c is shown in Fig. 8. As can be seen many cracks are subparallel to the direction of σ_1. The three-dimensional analysis revealed also that in general many triple junctions between cracks are observed, which gives some indication on the mean coordination number of the crack network: we will use this result later in the discussion. In Fig. 9, we present our results inferred from the mercury injection tests. This technique applied to crystalline rocks gives access to an equivalent crack aperture calculated using Washburn's equation. This parameter does not give access to the actual distribution of crack aperture, except for the so-called parallel model, an unrealistic case where all cracks are connected to the mercury reservoir but without connections between them [34]. Nevertheless, the distribution derived from the mercury injection tests can be used as a first estimate of the real distribution of crack apertures. Notice that the data were corrected from compressibility effects, as high pressures are applied on the mercury when tight rocks are tested. Doing so, we think that the quality of the data shown here is reasonably good. Only the results for the MC samples are presented in Fig. 9. There are two important features to highlight. First there is systematically a peak located at about 4–5 μm: without ruling out completely the possibility of a drawback in our experimental procedure, this feature could also be associated with a common background property of the intact granite. Unfortunately we could not check this argument. Second, another peak towards the lower aperture values is present, which is progressively shifted towards larger values, and widens up as the induced damage in the MC samples increases. We think that this part of the distribution is specifically associated with the stress-induced cracks generated during the mechanical tests. Interestingly, our results consistently show that the mean crack aperture and the variance of the distribution increase for more damaged samples. In other words, more heterogeneity is introduced during the mechanical cracking processes. The mercury injection data can also be used in models for the transport properties of the granite samples, as will be discussed later.

6. Discussion

Let us briefly summarize our results. We were successful in preparing a set of granite samples in which irreversible thermal or mechanical cracking was induced. For the latter, the key point was to record continuously the acoustic emission activity while loading the samples under high deviatoric stresses. In general, thermal and mechanical cracking have a strong influence on the acoustic and the transport properties. Ultrasonic velocities and Q factors for the cracked

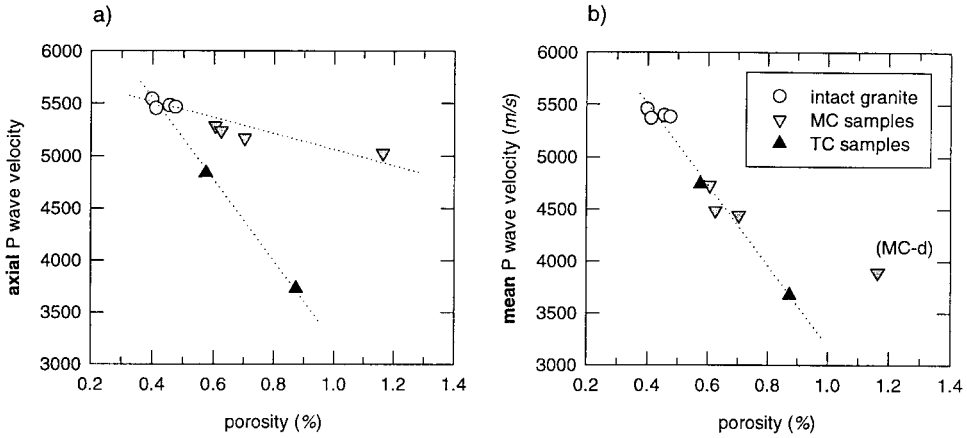


Fig. 10. Correlation between selected measured properties: (a) Axial *P*-wave velocity vs. porosity. (b) Mean *P*-wave velocity vs. porosity.

samples are significantly lower whereas permeability and electrical conductivity are higher than for the intact rock. In addition, good correlations were found in most of the cases with the amount of crack damage induced. The main difference between the MC and TC samples is the presence of a strong *P*-wave velocity anisotropy in the former (up to 45%) not observed in the latter (less than 4%). As showed by a thorough microstructure analysis, this is due to the presence of an anisotropic crack population which developed under a strongly anisotropic stress field during the mechanical compression tests, with cracks oriented preferentially in the direction of maximum stress (i.e. the core axis direction). In the following we will discuss the possibility to distinguish the effect of thermal cracking with respect to stress-induced cracking from correlations between different physical properties. Finally, theoretical models for the acoustic and transport properties of rocks will be used in order to interpret our experimental results in terms of crack properties.

6.1. Correlations between physical properties

Fig. 10a shows the correlation between porosity and axial *P*-wave velocity. Interestingly, it is possible to separate on this plot the MC samples from the TC samples. The velocity decrease with porosity is fairly linear. The slope for the thermal cracking set is very steep. As porosity is mainly associated with cracks, such a sharp decrease is due to the strong sensitivity of acoustic and mechanical properties to crack density and aspect ratio [40,41]. For the MC samples, the slope is weaker: this result is a consequence of the anisotropic crack distribution in these samples. As pointed out before, *P*-waves travelling in the axial direction are associated with compressional vibrations perpendicular to the preferentially oriented cracks: consequently, stress-induced cracks are expected to have little effect on axial *P*-wave velocity, whereas they contribute significantly in porosity enhancement. On the contrary, a stronger decrease in axial velocity is expected if the distribution of crack orientations is isotropic. This is so because statistically all crack orientations are equally represented. In Fig. 10b, we plotted the mean *P*-wave velocity (i.e. average between axial and radial velocity) as a function of porosity. Differences between mean and axial *P*-wave velocities are important only if anisotropy is significant. Interestingly, all the data follow now the same trend, except for sample MC-d. A single evolution is obtained because the effect of anisotropy has been averaged out: on such a plot, it is not possible to discriminate between thermal cracking and stress-induced cracking. Sample MC-d is clearly off the trend, which may be due to the different sample preparation technique used (water saturated sample tested under undrained conditions).

6.2. Estimation of microstructural parameters from velocity data

Taking advantage of our data set on *P*-wave velocities, it is possible to infer some important microstructural parameters like crack density and aspect ratio. Our analysis is based on Hudson's work [42] who calculated the velocity of elastic waves travelling through cracked materials. In the model, the cracks are represented by inclusions with a circular shape, dry or filled with water, randomly distributed in position. The crack aperture does not come into the calculations: in other words, this means that the model is suitable only for distributions of cracks with small aspect ratio (aperture$\ll$length). One assumption in the model is that the wavelength of the *P*-wave is much larger than the size of the inclusions: this condition is met in our experiments as long as the cracks are shorter than 5 mm. Both randomly oriented cracks or

aligned cracks can be considered in the model, which of course is ideal regarding the present study. The crack density parameter ε, as introduced by O'Connell and Budiansky [43] is defined by $\varepsilon = Na^3$ where N is the number density of cracks and a is the mean radius of the cracks. For a distribution of randomly oriented dry cracks, in the assumption of dilute crack concentration ($\varepsilon \ll 1$), the model allows to calculate the crack density parameter from the velocity V and density ρ of the cracked rock, and from the Lamé elastic constants λ_s and μ_s of the solid:

$$\varepsilon = [\lambda_s + 2\mu_s - \rho V^2]/\Gamma$$

where

$$\Gamma = \frac{4}{27}\frac{(3\lambda_s + 2\mu_s)^2(\lambda_s + 2\mu_s)}{\mu_s(\lambda_s + \mu_s)} + \frac{64}{135}\frac{\mu_s(\lambda_s + 2\mu_s)(9\lambda_s + 10\mu_s)}{(3\lambda_s + 4\mu_s)(\lambda_s + \mu_s)}. \quad (4)$$

For an anisotropic crack distribution, the model gives the expression for the velocity V_θ of a wave propagating in a direction defined by the angle θ with the direction of alignment of the crack axis

$$V_\theta^2 = V_{sol}^2\left\{1 - \frac{4}{3}\varepsilon\left(\frac{(\lambda_s + 2\mu_s\cos^2\theta)^2}{\mu_s(\lambda_s + \mu_s)} + \frac{16\mu_s}{3\lambda_s + 4\mu_s}\sin^2\theta\cos^2\theta\right)\right\}, \quad (5)$$

where V_{sol} is the P-wave velocity in the solid. In our case, we can apply this relation for the axial velocity ($\theta = 90°$) and for the radial velocity ($\theta = 0°$) measured on our granite samples. Therefore, two different values for the crack parameter can be derived, from which the statistical average is taken. Knowing the average mineralogical composition of La Peyratte granite [18], the elastic constants of the solid where calculated using the Voight–Reuss–Hill average [33]: we found $\lambda_s = 31.3$ GPa and $\mu_s = 31.2$ GPa. The average density of the solid being $\rho_s = 2660$ kg/m^3, the velocity in the solid is $V_{sol} = 5935$ m/s. Following O'Connell and Budiansky [43], the expression of the crack density ε can be calculated for oblate ellipsoids by

$$\varepsilon = \frac{3}{4\pi}\frac{\phi}{\alpha}, \quad (6)$$

where the aspect ratio α is defined as the maximum aperture divided by the crack diameter. Although the crack geometry is slightly different in both models, one can estimate the aspect ratio from Eq. (6) using the crack density parameter derived from Hudson's analysis (the smaller the crack aspect ratio, the better the approximation). Our results are reported in Table 3. For the intact granite and the TC samples, the crack density was calculated using Eq. (4), as no significant anisotropy was detected from our measurements. For the MC samples, ε was derived from Eq. (5) which is suitable for anisotropic cracked media. Notice that in our samples the normal of the crack planes are not aligned in the same direction like in Hudson's model: indeed, due to the axisymmetrical geometry of the loading, there is no constraint on the azimuthal orientation of the cracks. However, we think that we can still apply the model to get reasonable estimates for ε. As expected, the crack density parameter for the cracked samples is significantly higher than for the intact granite, and increases both within the TC and the MC series. It is important to realize that the ε values derived from Eq. (5) are specific to the anisotropic crack distribution (indeed, no other cracks are considered in Hudson's model). We know however that for La Peyratte granite there is also an isotropic population of cracks present in the intact rock: an estimate of the true crack density can be obtained by adding the value corresponding to the intact rock (Table 3). For both sets of samples, similar values were found for the crack density. Interestingly, these values are comparable to measured values reported for Westerly granite by Hadley [12]. Calculating the mean crack aspect ratio from Eq. (6), we did not find significant changes for this parameter, with values close to 10^{-2}. Interestingly, smaller values were obtained for the deformed samples compared to the intact granite, in agreement with Hadley's measurements on Westerly granite who found aspect ratios in the same range as ours. This means that the cracks induced in the TC and MC samples are thinner and/or longer, or that crack length increases more than the aperture. This last scenario seems to hold at least for the MC samples, as we know from the mercury injection data that the aperture distribution is shifted towards higher values when more cracks are induced. It may also be true for the TC samples, but we have no data to con-

Table 3
Inferred crack density, aspect ratio and permeability for deformed La Peyratte granite samples

	Crack density ε	Aspect ratio α	Permeability (10^{-21} m^2)
Intact	0.075 ± 0.004[a]	0.013 ± 0.002	–
220°C	0.139[a]	0.0098	–
450°C	0.245[a]	0.0085	–
MC-a	0.197[b]	0.0073	88 ± 9
MC-b	0.216[b]	0.0069	104 ± 6
MC-c	0.231[b]	0.0073	908 ± 57
MC-d	0.277[b]	0.0100	4169 ± 523

[a] Eq. (4).
[b] Eq. (5).

firm. Homand–Etienne and Houpert [5] observed in their thermal cracking experiments that the crack width was enhanced with increasing thermal damage, however they did not observe any significant variation in crack length. From our experience, investigating the crack length distribution on thin sections is not an easy task because it is often difficult to estimate the actual limits of a single crack which may also extend out of the investigation field.

Finally we made some numerical simulation of fluid flow in cracked systems, using the microstructural information available to estimate the permeability of the deformed and intact granite. To do so, we used the network model developed by David et al. [44]. The input data for the model are the rock porosity, the distribution of crack aperture estimated from the mercury injection data, the crack aspect ratio, and the crack network topology. For the latter, we chose a mean coordination number equal to 3, as we observed in many cases triple junctions between adjacent cracks on the CSLM images of reconstructed crack networks: therefore, a two-dimensional 51×50 hexagonal network model was used in the calculation [44]. Cracks are represented by channels with an elliptical cross-section defined by the aspect ratio α [45]: we allowed this last parameter to vary in a small range centered on the value given in Table 3. The results of the simulations are reported in Table 3 for the MC samples. In each case, 10 simulations were done corresponding to different network realizations of the same statistical crack distribution, in order to assign standard errors to the calculations. The results of the simulations are compared to the permeability data at the lowest effective pressure applied in our experiments (3 MPa). In general, the numerical simulations give the correct order of magnitude for the permeability, except for sample MC-d. This is a valuable result if one takes into account that permeability is a parameter with an extremely large range of variation. It also validates in first approximation the use of mercury injection data to estimate the actual aperture distribution. It is important to realize that the variation of the predicted permeability from sample to sample is a direct consequence of changes in the equivalent aperture distribution as shown in Fig. 9. Consequently, the abnormal behavior of sample MC-d (Fig. 7) obviously could not be reproduced, given the crack aperture spectrum in Fig. 9. The relative permeability increase from sample MC-a to MC-c is overestimated by the model. This is probably due to the fact that the network model used is oversimplified for several reasons: constant coordination number, two-dimensional topology, little control of crack aspect ratio and length statistics. Network modelling allows also to simulate the pressure dependence of permeability [45]. The pressure sensitivity of permeability (and other properties) depends mainly on the aspect ratio of the pore inclusions [40]: we found that aspect ratios of order 10^{-2} do not allow to reproduce the strong permeability drop as a function of effective pressure (Fig. 7). A simple calculation based on the analysis of Le Ravalec et al. [32] shows that our permeability data are consistent with a mean crack aspect ratio of order 10^{-4}. Obviously this is in strong disagreement with the prediction of Hudson's model based on *P*-wave velocity data, and with our fluid flow simulations. A similar conclusion was mentioned by Hadley [12] who pointed out that "aspect ratios in situ must be 2–3 orders of magnitude smaller than those observed or inferred in laboratory granite samples in order to account for observed stress sensitivities of shallow crustal seismic velocities".

7. Conclusion

In conclusion, we have presented new data on the influence of thermal cracking and stress-induced cracking on important physical rock properties: porosity, acoustic velocity, attenuation, electrical conductivity and permeability. All in all, good correlations were found between the evolution of these properties and the amount of damage induced. For the mechanical cracking process, our results clearly show that acoustic emissions recording is a powerful technique to control in real time the evolution of the cracking processes. For the MC samples, a strong anisotropy in the acoustic velocity data is associated to an anisotropic crack distribution with a preferred crack orientation parallel to σ_1, the direction of applied maximum stress. This was nicely confirmed by microstructure analyses and 3D reconstructions of crack networks from confocal scanning laser microscopy images. Crack parameters were inferred from Hudson's model [42] based on the propagation of elastic waves in cracked solids. Using these parameters along with results from mercury injection tests as input data in network models for fluid flow through cracked media [45], we found the same order of magnitude for the predicted permeability compared to experimental measurements. However the numerical simulations could not reproduce relative permeability variations due to oversimplified assumptions in the model.

How can the results presented in this paper be used in practical situations? Let us consider for example the simple fact that, according to our experimental study the permeability can be enhanced by more than two orders of magnitude for a damaged granite compared to an intact granite. This will obviously have dramatic consequences on the kinetics of fluid migration. In this regard, this is especially important to consider within the frame of safety assessment for potential nuclear waste storage in crystalline rock environment. Another

possible application is the survey of hazardous geological environments. Monitoring continuously specific physical properties such as acoustic velocities or attenuation can help to characterize the evolution of the rock's physical state provided that there is a good understanding of the link between rock properties and microstructures. Moreover, if specific attributes can be recognized, such as isotropic or anisotropic behavior, the origin of the damaging process may be identified. The evaluation of seismic or volcanic hazards is obviously concerned by these issues. The data set presented in this paper gives some valuable insight into the interplay between the evolution of rock microstructures and variations of physical properties in granites, as a consequence of either thermal or tectonic loading. In this regard, our study can help to bring some answers to these important questions.

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